

Theory

Gaussian Beam Optics

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Abstract

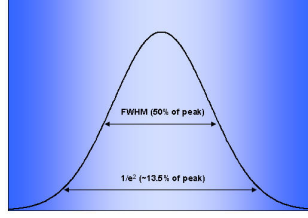
The purpose of the document is to work through the theory and experimental design for optimizing second-harmonic generation (SHG) at 1064 nm using a periodically poled lithium niobate (PPLN) crystal ($50 \times 2 \times 0.5 \text{ mm}^3$). In this document, I define Gaussian beam parameters, including beam waist, Rayleigh range, divergence, and confocal parameter, in order to determine the optimal focusing geometry. The Boyd-Kleinman condition for maximum CW SHG efficiency ($L/b = 2.84$) is a target focused waist of approximately $54.6 \text{ }\mu\text{m}$, but experimental constraints redefines a target of $77\text{--}80 \text{ }\mu\text{m}$ ($L/b \approx 1.3 - 1.4$). Starting from a collimated 0.56 mm beam at 1064 nm with a measured divergence of 1.21 mRad , the collimator waist is estimated to be $\omega_{01} \approx 280 \text{ }\mu\text{m}$, located 6.2 mm behind the collimator face. That being said, the calculated waist radius of roughly 0.279 mm would result in a beam larger than the crystal itself can accommodate. To account for this, I used an ABCD ray-transfer matrix to propagate a beam, constructing two equations relating a chosen lens focal length f to the required lens position d_1 and image distance d_2 , with magnification $\alpha \approx 0.286$.

1 Experiment Specifics

1. Wavelength: 1064 nm
2. Collimator output beam size: $0.56 \text{ mm } \frac{1}{e^2} \text{ diameter}$
3. PPLN crystal: $50 \times 2 \times 0.5 \text{ mm}^3$
4. Previous best measured output: $57 \text{ mW green, PPLN at } 59.5^\circ\text{C}$
5. Previous calculated waist: about $45 \text{ }\mu\text{m}$
6. Desired next target: about $77\text{--}80 \text{ }\mu\text{m}$

2 Definitions and Equations

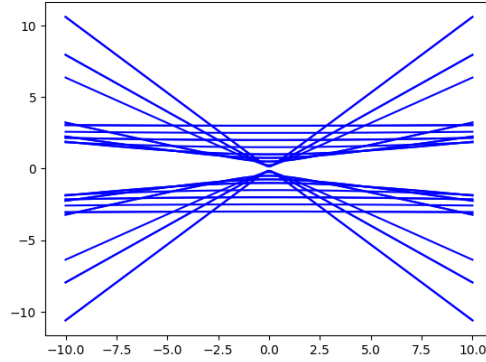
Beam Diameter ($\frac{1}{e^2}$): The $\frac{1}{e^2}$ beam diameter captures the diameter where the intensity of the beam drops to 13.53%; in other words, 86.47% of the beam intensity lies within the given diameter.



Beam Waist (ω_0): The beam waist is the portion of the beam where the radius is minimized, denoted ω_0 . The tighter the focus the smaller the waist, but this introduces a large amount of divergence, as seen in the z -dependent beam width equation

$$\omega^2(z) = \omega_0^2 \left[1 + \left(\frac{\lambda z}{\pi \omega_0^2} \right)^2 \right] \quad (1)$$

Small waists are achieved by a “short focal length and a high numerical aperture largely filled by the input beam.” [7]



Radius of Curvature (R_g): The radius of curvature is defined by Eq. 2. At its boundaries, $\lim_{z \rightarrow 0} R_g(z) = \infty$ where the wavefront is perfectly flat and acts like a section of a circle with infinite radius, and $\lim_{z \rightarrow \infty} R_g(z) = z$ where the wavefront averages out to a radius equal to the distance from the waist.

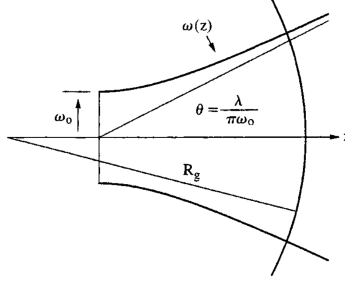
$$R_g(z) = z \left[1 + \left(\frac{\pi \omega_0^2}{\lambda z} \right)^2 \right] \quad (2)$$

Rayleigh Range (z_0): The Rayleigh range is the distance at which the cross-sectional area of the beam is no more than twice that of the beam waist, ensuring at least 50% of the beam is retained.[8]

$$z_0 = \frac{\pi \omega_0^2}{\lambda} \quad (3)$$

Beam Divergence (θ): The beam divergence is the slope of the linear component of beam expansion (the far-field expansion), measured as a half-angle dependent on the wavelength and inversely on the waist:

$$\theta = \frac{\lambda}{\pi \omega_0} \quad (4)$$



Confocal Parameter (b): The confocal parameter uses the Rayleigh range to quantify the useful depth of focus—it is “the total length of the focused region around the beam waist.” [8]

Boyd–Kleinman Focusing: $h = 1.068$ at $L/b = 2.84$ [1]; $h = 1.066$ at $L/b = 2.81$ [4]. Here h is the conversion efficiency, L is the crystal length, and $b = 2z_0$ is the confocal parameter. We work backwards from these values to find the optimal waist; see Section 3.2.

1

2

3 Calculations

3.1 Initial Beam Parameters

Starting from a $0.56 \text{ mm } 1/e^2$ beam diameter at 1064 nm , what are the approximate Gaussian beam parameters?

Using Eq. 3:³

$$z_{01} = \frac{\pi \omega_{01}^2}{\lambda} = \frac{\pi \cdot (0.00028)^2}{1.064 \times 10^{-6}} \approx 0.23149 \text{ m} \approx 23.15 \text{ cm}$$

This results in $L/b = 0.108$, far from the Boyd–Kleinman optimum.

Using Eq. 4:

$$\theta = \frac{\lambda}{\pi \omega_{01}} = \frac{1.064 \times 10^{-6}}{\pi \cdot 0.00028} \approx 1.210 \times 10^{-3} \text{ rad}$$

3.2 Boyd–Kleinman Focusing Targets

3.2.1 Optimum: $L/b = 2.84$ [1]

Given $L/b = 2.84$ and $L = 50 \text{ mm}$.

$$b = \frac{L}{2.84} = \frac{0.05}{2.84} \text{ m}$$

Using Eq. 3:

$$\frac{0.05}{2.84} = b = 2z_{02} = \frac{2\pi \omega_{02}^2}{\lambda} \Rightarrow \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot 2.84}} \approx 54.602 \text{ } \mu\text{m}$$

3.2.2 Optimum: $L/b = 2.81$ [4]

Given $L/b = 2.81$ and $L = 50 \text{ mm}$.

$$b = \frac{L}{2.81} = \frac{0.05}{2.81} \text{ m}$$

Using Eq. 3:

$$\frac{0.05}{2.81} = b = 2z_{02} = \frac{2\pi\omega_{02}^2}{\lambda} \Rightarrow \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot 2.81}} \approx 54.892 \text{ } \mu\text{m}$$

3.2.3 Conservative rule of thumb: $L/b = 1$ [2]

4

Given $L/b = 1$ and $L = 50 \text{ mm}$.

$$b = L = 0.05 \text{ m}$$

Using Eq. 3:

$$0.05 = b = 2z_{02} = \frac{2\pi\omega_{02}^2}{\lambda} \Rightarrow \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi}} \approx 92.017 \text{ } \mu\text{m}$$

3.2.4 Working Backwards from Target Waists

To recover a desired ω_{02} , we solve for $\xi \equiv L/b$:

1. $\omega_{02} = 45 \text{ } \mu\text{m}$:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (45 \times 10^{-6})^2} \approx 4.181$$

This ratio is too large (waist too tight), leading to excessive diffraction and reduced efficiency.

2. $\omega_{02} = 77 \text{ } \mu\text{m}$:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (77 \times 10^{-6})^2} \approx 1.428$$

Within the proper range.

3. $\omega_{02} = 80 \text{ } \mu\text{m}$:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (80 \times 10^{-6})^2} \approx 1.323$$

Within the proper range.

3.3 Collimator Waist Estimation

*Repeating the analysis without assuming the beam waist is at the collimator face.*⁵

1. Using Eq. 4 to recover the waist from the measured divergence:

$$\omega_{01} = \frac{\lambda}{\pi\theta} = \frac{1.064 \times 10^{-6}}{\pi \cdot 0.00121} \approx 279.9 \text{ } \mu\text{m}$$

2. Using the beam propagation equation (Eq. 5, from Eq. 2-27 in [5]):

$$\omega^2(z) = \omega_{01}^2 \left[1 + \left(\frac{\lambda z}{\pi\omega_{01}^2} \right)^2 \right] \quad (5)$$

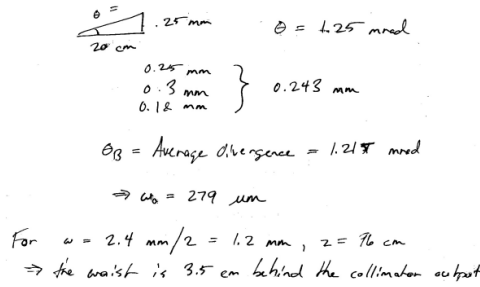
Solving for z :

$$\frac{\omega^2(z)}{\omega_{01}^2} - 1 = \left(\frac{\lambda z}{\pi \omega_{01}^2} \right)^2 \Rightarrow z = \pm \left(\sqrt{\frac{\omega^2(z)}{\omega_{01}^2} - 1} \right) \frac{\pi \omega_{01}^2}{\lambda}$$

Setting $\omega(z) = 0.00028$ m (the $1/e^2$ radius at the collimator aperture):

$$z \approx \pm \left(\sqrt{\frac{(0.00028)^2}{(0.0002799)^2} - 1} \right) \frac{\pi \cdot (0.0002799)^2}{1.064 \times 10^{-6}} \approx \pm 6.184 \text{ mm}$$

The waist is located approximately 6.2 mm *behind* the collimator face.⁶



3. Recalculating the Rayleigh range with the corrected waist:

$$z_{01} = \frac{\pi \omega_{01}^2}{\lambda} = \frac{\pi \cdot (2.799 \times 10^{-4})^2}{1.064 \times 10^{-6}} \approx 23.132 \text{ cm}$$

4. Because the beam waist $\omega_{01} \approx 280 \mu\text{m}$ is the *minimum* radius of the laser, the beam can only diverge from that point and would “scrape” the inside of the PPLN crystal. A focusing lens is therefore required.

3.4 Lens Selection

1. The magnification relation from [3]:

$$\omega_{02} = \alpha \cdot \omega_{01} \Rightarrow \alpha = \frac{\omega_{02}}{\omega_{01}} \quad (6)$$

2. The lens-placement formula from [3]:

$$\alpha = \frac{f}{\sqrt{(|d_1| - f)^2 + z_{01}^2}} \quad (7)$$

Solving for $|d_1|$:

$$\alpha^2 [(|d_1| - f)^2 + z_{01}^2] = f^2 \Rightarrow |d_1| = \sqrt{\left(\frac{f}{\alpha} \right)^2 - z_{01}^2} + f$$

3. Image-distance formula from [3]:

$$d_2 = f + \alpha^2 (d_1 - f) \quad (8)$$

4. **Numerical values.** With $\omega_{02} = 80 \mu\text{m}$ and $\omega_{01} = 279.9 \mu\text{m}$:

$$\alpha = \frac{80}{279.9} \approx 0.28582, \quad z_{01} \approx 0.23132 \text{ m}$$

For a focal length f chosen in the lab:

$$d_1 = \sqrt{\left(\frac{f}{0.28582}\right)^2 - (0.23132)^2} + f, \quad d_2 = f + (0.28582)^2(d_1 - f)$$

5. **Derivation of equations used above** Starting from [5]:

$$\frac{1}{q_1} = \frac{1}{R_g} - i \frac{\lambda}{\pi \omega_{01}^2} \quad (9)$$

where R_g is the radius of curvature of the input beam. Taking the limit $R_g \rightarrow \infty$ (collimated input):

$$q_1 = i \frac{\pi \omega_{01}^2}{\lambda} = iz_{01}$$

The ABCD ray-transfer matrix for free-propagation d_1 , thin lens f , then free-propagation d_2 is:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & d_2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{pmatrix} \begin{pmatrix} 1 & d_1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{d_2}{f} & d_1 + d_2 - \frac{d_1 d_2}{f} \\ -\frac{1}{f} & 1 - \frac{d_1}{f} \end{pmatrix}$$

so $A = 1 - \frac{d_2}{f}$, $B = d_1 + d_2 - \frac{d_1 d_2}{f}$, $C = -\frac{1}{f}$, $D = 1 - \frac{d_1}{f}$.

The q -parameter transforms as (Eq. 10 from [5]):

$$q_2 = \frac{Aq_1 + B}{Cq_1 + D} \quad (10)$$

Substituting $q_1 = iz_{01}$:

$$q_2 = \frac{\left(1 - \frac{d_2}{f}\right) iz_{01} + \left(d_1 + d_2 - \frac{d_1 d_2}{f}\right)}{\left(-\frac{1}{f}\right) iz_{01} + \left(1 - \frac{d_1}{f}\right)}$$

Multiplying numerator and denominator by the complex conjugate of the denominator, $\left(1 - \frac{d_1}{f}\right) + i\left(\frac{z_{01}}{f}\right)$:

Real part:

$$\begin{aligned} \text{Re}(q_2) &= \frac{-\left(1 - \frac{d_2}{f}\right) \left(\frac{z_{01}^2}{f}\right) + \left(d_1 + d_2 - \frac{d_1 d_2}{f}\right) \left(1 - \frac{d_1}{f}\right)}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \\ &= \frac{d_2 \left[\left(1 - \frac{d_1}{f}\right)^2 + \frac{z_{01}^2}{f^2}\right] + d_1 \left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \\ &= d_2 + \frac{d_1 \left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \end{aligned}$$

Imaginary part:

$$\text{Im}(q_2) = \frac{z_{01} \left[\left(1 - \frac{d_1}{f} - \frac{d_2}{f} + \frac{d_1 d_2}{f^2}\right) + \left(\frac{d_1}{f} + \frac{d_2}{f} - \frac{d_1 d_2}{f^2}\right) \right]}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} = \frac{z_{01}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2}$$

Full expression:

$$q_2 = \left[d_2 + \frac{d_1 \left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \right] + i \left[\frac{z_{01}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \right]$$

Setting the real part to zero to satisfy the waist condition: $\lim_{z \rightarrow 0} \frac{1}{R_g(z)} = 0$ results in

$$d_2 = \frac{f[(d_1 - f)^2 + z_{01}^2] + f^2(d_1 - f)}{(d_1 - f)^2 + z_{01}^2} = f + \frac{f^2(d_1 - f)}{(d_1 - f)^2 + z_{01}^2}$$

$$d_2 = f + \alpha^2(d_1 - f)$$

proving Eq. 8, where $\alpha^2 = \frac{f^2}{(d_1 - f)^2 + z_{01}^2}$ follows Eq. 7.

4 Use Case

PPLN:⁷ To get the most out of our PPLN crystals, four key aspects must be considered:

1. Crystal Length

- (a) No special action required at this stage.

2. Polarization

- (a) The input light must be e-polarized, i.e. the polarization must be aligned with the dipole moment of the crystal. This is accomplished by aligning the polarization axis of the light parallel to the thickness of the crystal.[2]

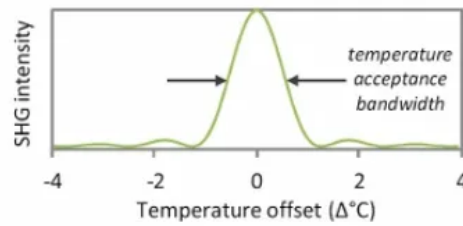
3. Focusing and the Optical Arrangement

- (a) For SHG with CW lasers, the Boyd–Kleinman result shows that optimum conversion efficiency is achieved when $L/b = 2.84$. [2]

4. Temperature and Period

- (a) The crystal is manufactured with periodic stripes of alternating polarity with a spacing called the poling period. This quasi-phase-matched arrangement allows the harmonic power to build up continuously.
- (b) The effective poling period can be tuned by changing the crystal temperature, which causes thermal expansion or contraction.
- (c) Note: The refractive index of the crystal also changes with temperature.

- (d) The output intensity follows a sinc^2 curve (shown below), defining the temperature acceptance bandwidth.



- (e) Acceptance bandwidth is inversely proportional to crystal length.
- (f) The optimum temperature can be found by heating the crystal to 20°C above the calculated phase-matching temperature, then allowing it to cool while monitoring the generated output power.[2]

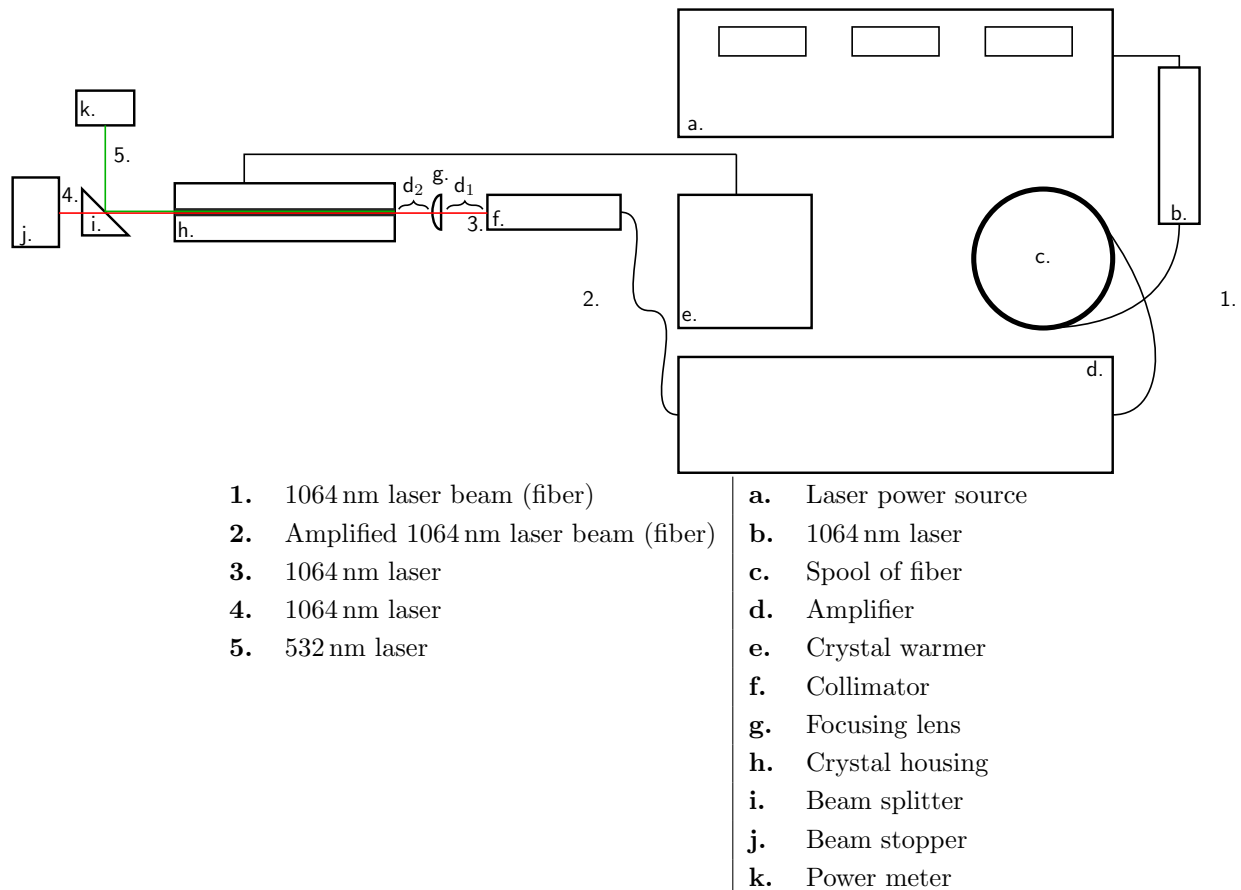


Figure 1: Experimental setup and component legend.

Signal Processing Calculations

This section was taken from the Jupyter Notebook I used to calculate the beam params. The purpose of this notebook is to work through the theory and experimental design for optimizing second-harmonic generation (SHG) at 1064 nm using a periodically poled lithium niobate (PPLN) crystal ($50 \times 2 \times 0.5 \text{ mm}^3$). We define Gaussian beam parameters—including beam waist, Rayleigh range, divergence, and confocal parameter—to determine the optimal focusing geometry.

The Boyd-Kleinman condition for maximum CW SHG efficiency ($L/b = 2.84$) sets a target focused waist of approximately $54.6 \text{ }\mu\text{m}$, but experimental constraints redefine a target of $77\text{--}80 \text{ }\mu\text{m}$ ($L/b \approx 1.3 - 1.4$).

Starting from a collimated 0.56 mm beam at 1064 nm with a measured divergence of 1.21 mRad , the collimator waist is estimated to be $\omega_{01} \approx 280 \text{ }\mu\text{m}$, located 6.2 mm behind the collimator face. Because a waist radius of roughly 0.279 mm would result in a beam larger than the crystal itself can accommodate, we use an ABCD ray-transfer matrix to propagate the beam, constructing equations relating a chosen lens focal length f to the required lens position d_1 and image distance d_2 . **Test 1:**

- Wavelength: 1064 nm
- Collimator output beam size: 0.56 mm ($1/e^2$ diameter)
- PPLN crystal: $50 \times 2 \times 0.5 \text{ mm}^3$
- Previous best measured output: 57 mW green, PPLN at 59.5°C
- Previous calculated waist: $45 \text{ }\mu\text{m}$
- **Desired next target:** $77\text{--}80 \text{ }\mu\text{m}$

```
1 import numpy as np
2 import math
3 wavelength = 1064e-9 # m
4 collimator_diam = 0.56e-3 # m
5 crystal_length = 50e-3 # m
6 measured_divergence = 1.21e-3 # rad
```

Repeating the analysis without assuming the beam waist is exactly at the collimator face (using the measured 1.21 mRad divergence).

1. Recover the true waist from the measured divergence.
2. Calculate the distance z from the true waist to the collimator face where the beam radius is 0.28 mm .
3. Recalculate the true Rayleigh range.

```
1 # 1. True initial waist from measured divergence
2 omega_01 = wavelength / (np.pi * measured_divergence)
3 print(f"True Collimator Waist (omega_01): {omega_01 * 1e6:.1f} um")
4
5 # 2. Distance from waist to collimator face
6 omega_aperture = collimator_diam / 2
7 # Solving the omega(z) equation for z
8 z_collimator = np.sqrt((omega_aperture**2 / omega_01**2) - 1) * (np.pi * omega_01
    **2) / wavelength
```

```

9 print(f"Distance from waist to face: {z_collimator * 1e3:.2f} mm")
10
11 # 3. True Rayleigh range of the initial beam
12 z_01 = (np.pi * omega_01**2) / wavelength
13 print(f"Initial Rayleigh Range (z_01): {z_01 * 100:.2f} cm")

```

True Collimator Waist (omega_01): 279.9 um
Distance from waist to face: 6.11 mm
Initial Rayleigh Range (z_01): 23.13 cm

```

1 targets_um = [45, 54.6, 77, 80] # um
2
3 print("Boyd-Kleinman Ratios (L/b) for target waists:")
4 print("-" * 50)
5 for target in targets_um:
6     omega_02 = target * 1e-6
7     b = (2 * np.pi * omega_02**2) / wavelength
8     xi = crystal_length / b
9     print(f"Target Waist: {target:>4.1f} um | L/b = {xi:.3f}")

```

Boyd-Kleinman Ratios (L/b) for target waists:

Target Waist: 45.0 um | L/b = 4.181
Target Waist: 54.6 um | L/b = 2.840
Target Waist: 77.0 um | L/b = 1.428
Target Waist: 80.0 um | L/b = 1.323

```

1
2 target_omega_02 = 80e-6 # 80 um
3 focal_length = float(input("Enter the lens focal length (in mm): ")) * 1e-3
4
5 # Calculate Magnification
6 alpha = target_omega_02 / omega_01
7
8 # Calculate d1 (Distance from initial waist to lens)
9 inside_sqrt = (focal_length / alpha)**2 - z_01**2
10
11 if inside_sqrt >= 0:
12     d1 = np.sqrt(inside_sqrt) + focal_length
13     # Calculate d2 (Distance from lens to new focus inside crystal)
14     d2 = focal_length + (alpha**2) * (d1 - focal_length)
15
16     # Calculate physical distance from the collimator face to the lens
17     # Note: z_collimator was calculated in Cell 6 as a positive distance
18     d1_face = d1 - z_collimator
19
20     print(f"Magnification (alpha): {alpha:.4f}")
21     print(f"Lens focal length: {focal_length * 1000:.1f} mm")
22     print("-" * 60)
23     print(f"Optical Distance (Waist to Lens, d1): {d1 * 1000:.2f} mm")
24     print(f"--> Physical Distance (Collimator FACE to Lens): {d1_face * 1000:.2f} mm")
25     print("-" * 60)
26     print(f"Distance from Lens to Focus (d2): {d2 * 1000:.2f} mm")
27 else:
28     print(f"No valid real solution for a {focal_length * 1000:.1f} mm lens.")

```

Key: ω_{01} : Initial Waist | ω_{02} : Focused Waist | z_{01} : Initial Rayleigh Range | α : Magnification | f : Focal Length
 d_1 : Dist. to Lens | d_2 : Dist. to Focus

True Collimator Waist (ω_{01}): 279.9 μm
Distance from waist to face: 6.11 mm
Initial Rayleigh Range (z_{01}): 23.13 cm
Boyd-Kleinman Ratios (L/b) for target waists:

Target Waist: 45.0 μm | $L/b = 4.181$
Target Waist: 54.6 μm | $L/b = 2.840$
Target Waist: 77.0 μm | $L/b = 1.428$
Target Waist: 80.0 μm | $L/b = 1.323$
Magnification (α): 0.2858
Lens focal length: 88.3 mm

Optical Distance (Waist to Lens, d_1): 293.08 mm
--> Physical Distance (Collimator FACE to Lens): 286.96 mm

Distance from Lens to Focus (d_2): 105.03 mm

Test 2: Changes in code:

```
1 crystal_length = 25e-3
```

True Collimator Waist (ω_{01}): 279.9 μm
Distance from waist to face: 6.11 mm
Initial Rayleigh Range (z_{01}): 23.13 cm
Boyd-Kleinman Ratios (L/b) for target waists:

Target Waist: 38.8 μm | $L/b = 2.810$
Target Waist: 54.6 μm | $L/b = 1.420$
Target Waist: 65.1 μm | $L/b = 1.000$
Magnification (α): 0.1387
Lens focal length: 50.0 mm

Optical Distance (Waist to Lens, d_1): 326.57 mm
--> Physical Distance (Collimator FACE to Lens): 320.46 mm

Distance from Lens to Focus (d_2): 55.32 mm
Target Waist: 38.8 μm
Initial Waist: 279.9 μm
Magnification (α): 0.1387

Lens Name	f (mm)	d1 (mm)	d2 (mm)	Total Path (mm)
Lens 0.5Dia (35mm)	35.0	135.95	36.94	172.89
Lens 0.5Dia (40mm)	40.0	212.31	43.31	255.62
Lens 0.5Dia (50mm)	50.0	326.57	55.32	381.89
Lens 0.5Dia (60mm)	60.0	425.64	67.03	492.67
Lens 0.5Dia (75mm)	75.0	563.87	84.40	648.27
Lens 0.5Dia (100mm)	100.0	783.01	113.13	896.14

Key: ω_{01} : Initial Waist | ω_{02} : Focused Waist | z_{01} : Initial Rayleigh Range | α : Magnification | f: Focal Length
 d_1 : Dist. to Lens | d_2 : Dist. to Focus

Lens 0.5Dia (125mm)	125.0	996.21	141.75	1137.96
Lens 0.5Dia (150mm)	150.0	1206.65	170.32	1376.97
Lens 0.5Dia (175mm)	175.0	1415.58	198.86	1614.43
Lens 0.5Dia (200mm)	200.0	1623.57	227.38	1850.94
Lens 0.5Dia (400mm)	400.0	3275.18	455.29	3730.48
Lens 0.5Dia (500mm)	500.0	4098.17	569.19	4667.36

5 Experimental Results

5.1 Lab Methods

For the third test, I used an non particularly straight forward aproach. Due to teh limitations of the oiwer meter , I one wishes to manulally record specific measurements of both sensors, to trackand attemot to midigate power fluctuations in the laser, one can only sift back through recorded data as there soes not seem to eb a screen mide ti view in time measuremnts with an amount of meaningful precision. To get around this I started the power meter recording and a stopwatch, precise up to 10 ms at the same time. Then I would record lap times for each micron, letting the meter settle for roughly 40s ever time it would have issuse with my shirt, spiking the readings. I then went mannually through the 660,000 data points, and extrsacted all useable data, matching the lap times to the 10micro second intervals of the recording.

5.2 Output Power vs. Crystal Temperature

[Add graph here]

5.3 Achieved Waist Measurement

To determine the beam waist, two razor-blade profiling tests were conducted. The raw spatial derivatives of the power measurements were smoothed using various algorithms (Moving Average, Exponential Smoothing, Savitzky-Golay, Median Filter, and FFT-based denoising) and fit to a Gaussian beam profile. See Section 5.5 for a visual step-by-step breakdown of the signal processing and denoising pipeline.

6/26 Razor Test 1

Experimental Setup:

- Collimator face to reflection off wedge: 95 mm
- Wedge to mirror: 135 mm
- Mirror to lens: 142 mm
- Lens to blade: 66 mm
- Blade to sensor: 200 mm
- Focal length: 50 mm (1 in diameter)
- **Measured Distances:** $d_1 = 372$ mm, $d_2 = 66$ mm. Total Path: 438 mm.

Smoothing Method	ω ($1/e^2$ radius, μm)	Center (mm)
Moving Average	88.79 ± 3.13	12.1824
Exp. Smoothing	87.91 ± 5.16	12.1854
Savitzky-Golay	88.25 ± 4.20	12.1824
Median Filter	89.18 ± 8.84	12.1824
FFT-denoised	111.99 ± 0.54	12.1821

Table 1: Gaussian fit results for Razor Test 1.

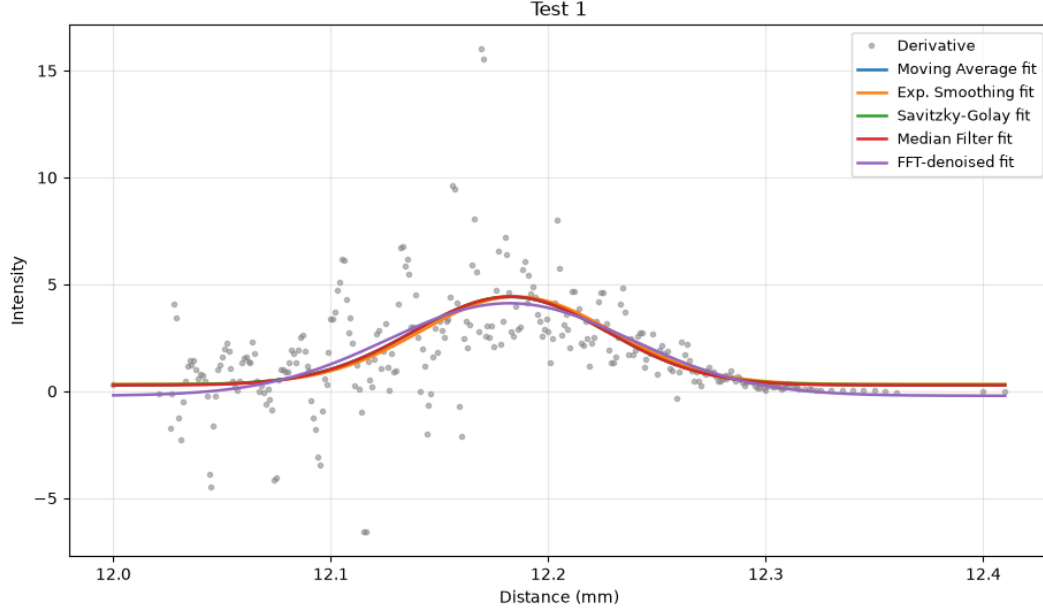


Figure 2: Final beam profile and Gaussian fits for Test 1.

6/29 Razor Test 2

Experimental Setup:

- Collimator face to reflection off wedge: 85 mm
- Wedge to mirror: 133 mm
- Mirror to lens: 102.5 mm
- Lens to blade: 56 mm
- Blade to sensor: 200 mm
- Focal length: 50 mm (1 in diameter)
- **Measured Distances:** $d_1 = 320.5$ mm, $d_2 = 56$ mm. Total Path: 376.5 mm.

Smoothing Method	ω ($1/e^2$ radius, μm)	Center (mm)
Moving Average	43.91 ± 0.76	5.1968
Exp. Smoothing	43.19 ± 1.13	5.2001
Savitzky-Golay	44.56 ± 0.94	5.1964
Median Filter	41.69 ± 2.83	5.1975
FFT-denoised	41.69 ± 0.57	5.1976

Table 2: Gaussian fit results for Razor Test 2.

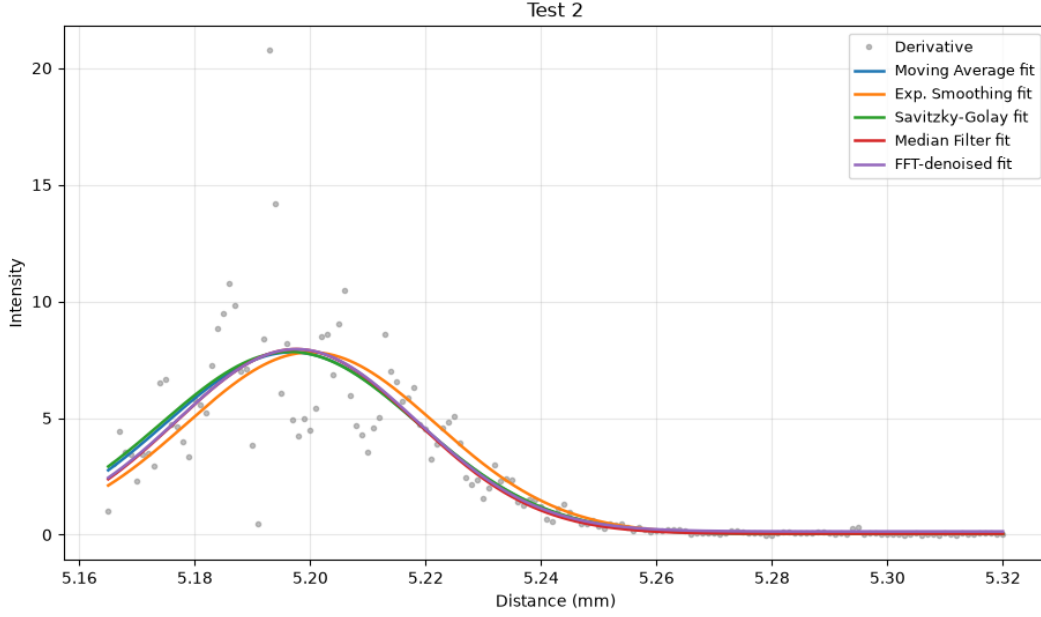


Figure 3: Final beam profile and Gaussian fits for Test 2.

6/29 Razor Test 3

Experimental Setup:

- Collimator face to reflection off wedge: 85 mm
- Wedge to mirror: 133 mm
- Mirror to lens: 102.5 mm
- Lens to blade: 56 mm
- Blade to sensor: 200 mm
- Focal length: 50 mm (1 in diameter)
- **Measured Distances:** $d_1 = 320.5$ mm, $d_2 = 56$ mm. Total Path: 376.5 mm.

Smoothing Method	ω ($1/e^2$ radius, μm)	Center (mm)
Moving Average	46.61 ± 1.52	0.0973
Exp. Smoothing	44.70 ± 2.49	0.0999
Savitzky-Golay	45.62 ± 2.00	0.0970
Median Filter	46.57 ± 4.83	0.0972
FFT-denoised	30.53 ± 1.50	0.0956

Table 3: Gaussian fit results for Razor Test 3.

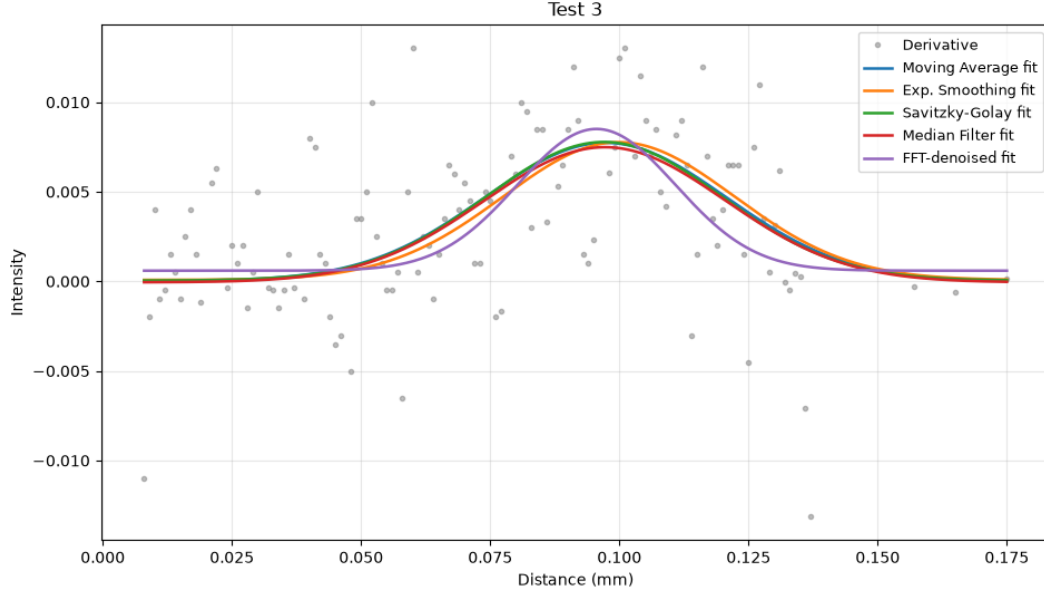


Figure 4: Final beam profile and Gaussian fits for Test 3.

Smoothing Method [Filtered]	ω ($1/e^2$ radius, μm)	Center (mm)
Moving Average	46.09 ± 1.38	0.0974
Exp. Smoothing	44.26 ± 2.20	0.1001
Savitzky-Golay	44.11 ± 2.25	0.0973
Median Filter	46.63 ± 4.32	0.0976
FFT-denoised	28.40 ± 1.90	0.0954

Table 4: Gaussian fit results for Razor Test 3 Filtered.

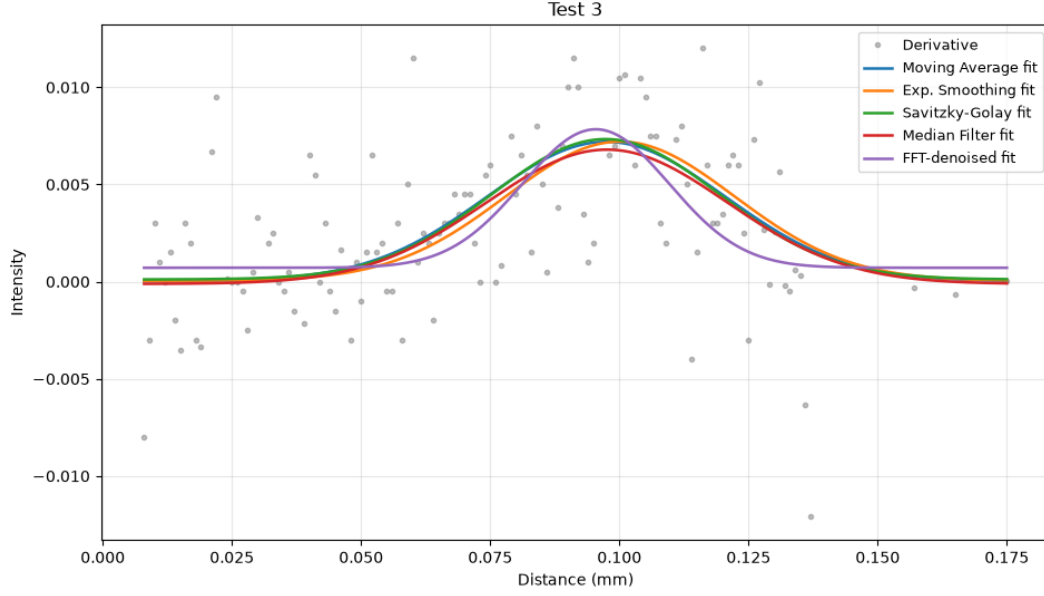


Figure 5: Final beam profile and Gaussian fits for Test 3 Filtered.

5.4 Comparison with Boyd–Kleinman Prediction

5.5 Data Processing and Analysis Code

The following Python code details the algorithms used to process the spatial beam profiling data. The algorithms include simple moving averages, exponential smoothing, Savitzky-Golay filtering, median filtering, and an FFT-based denoising approach.

```
1 import pandas as pd
2 import numpy as np
3 from scipy.optimize import curve_fit
4 from scipy import signal
5 from scipy.signal import savgol_filter, medfilt
6
7 # 1. Moving Average Filter
8 def moving_average_filter(signal, window=15):
9     kernel = np.ones(window) / window
10    pad = window // 2
11    padded = np.pad(signal, pad, mode='edge')
12    smoothed = np.convolve(padded, kernel, mode='same')
13    return smoothed[pad:pad + len(signal)]
14
15 # 2. Exponential Smoothing (EWMA)
16 def exponential_smooth(signal, alpha=0.25):
17     smoothed = pd.Series(signal).ewm(alpha=alpha, adjust=False).mean().values
18     return smoothed, alpha
19
20 # 3. Savitzky-Golay Filter
21 def sg_smooth(signal, window_length=25, polyorder=3):
22     if window_length % 2 == 0:
23         window_length += 1
24     return savgol_filter(signal, window_length=window_length, polyorder=polyorder)
25
26 # 4. Median Filter
27 def median_smooth(signal, window=9):
28     if window % 2 == 0:
29         window += 1
30     return medfilt(signal, kernel_size=window)
31
32 # 5. FFT-based denoising
33 def sigma_lf_fwhm(freqs, magnitude):
34     center_index = len(freqs) // 2
35     peak = magnitude[center_index]
36     half_max = peak / 2
37     right_index = center_index
38     while right_index < len(freqs) - 1 and magnitude[right_index] > half_max:
39         right_index += 1
40     if right_index > center_index:
41         f0, f1 = freqs[right_index - 1], freqs[right_index]
42         m0, m1 = magnitude[right_index - 1], magnitude[right_index]
43         f_half = f0 + (half_max - m0) * (f1 - f0) / (m1 - m0) if m1 != m0 else f1
44     else:
45         f_half = freqs[right_index]
46     fwhm = 2 * abs(f_half - freqs[center_index])
47     return fwhm / 2.355 # FWHM -> sigma for a Gaussian
48
49 def fft_denoise_profile(x_raw, profile_raw):
50     n = len(x_raw)
51     x_uniform = np.linspace(x_raw.min(), x_raw.max(), n)
52     p_uniform = np.interp(x_uniform, x_raw, profile_raw)
53     dx = x_uniform[1] - x_uniform[0]
```

```

54     spectrum = np.fft.fftshift(np.fft.fft(p_uniform))
55     freqs = np.fft.fftshift(np.fft.fftfreq(n, d=dx))
56     magnitude = np.abs(spectrum)
57
58     sigma_lf = sigma_lf_fwhm(freqs, magnitude)
59     lf_mask = np.abs(freqs) <= 4 * sigma_lf
60     spectrum_filtered = spectrum * lf_mask
61
62     hf_magnatudes = magnitude[~lf_mask]
63     sigma_hf = np.std(hf_magnatudes)
64     N = len(hf_magnatudes)
65     threshold = 3 * sigma_hf * np.sqrt(2 * np.log10(N))
66     spectrum_filtered[np.abs(spectrum_filtered) < threshold] = 0
67
68     denoised_uniform = np.real(np.fft.ifft(np.fft.ifftshift(spectrum_filtered)))
69     denoised = np.interp(x_raw, x_uniform, denoised_uniform)
70     return denoised, sigma_lf, freqs, magnitude, lf_mask
71
72
73 # Gaussian Beam Fit
74 def gaussian_beam(x, A, x0, w, offset):
75     return A * np.exp(-2 * (x - x0) ** 2 / w ** 2) + offset
76
77 def fit_beam_radius(x, profile):
78     center_guess = np.average(x, weights=profile)
79     variance_guess = np.average((x - center_guess)**2, weights=profile)
80     std_guess = np.sqrt(variance_guess)
81     w_guess = 2 * std_guess
82     p0 = [profile.max(), center_guess, w_guess, 0]
83     popt, pcov = curve_fit(gaussian_beam, x, profile, p0=p0)
84     perr = np.sqrt(np.diag(pcov))
85     return popt, perr

```

Listing 1: Smoothing and FFT Denoising Methodology

Signal Processing

The following figures provide additional visualization and help show how the data was processed.

Test 1:

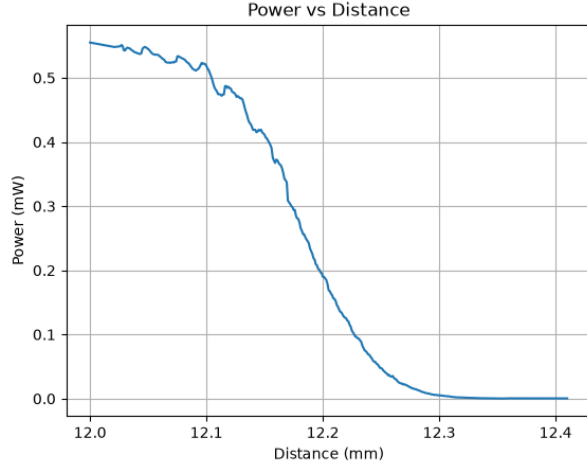


Figure 6: Test 1: Raw Power vs. Distance.

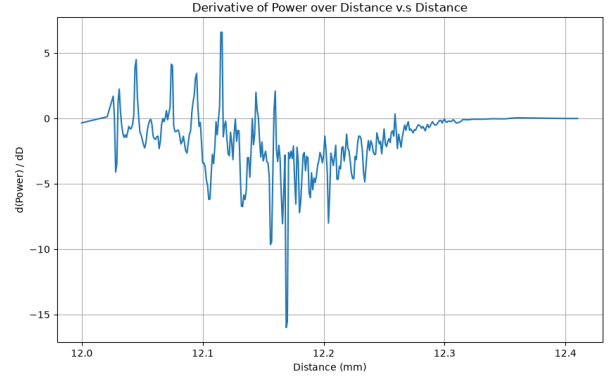


Figure 7: Test 1: Raw spatial derivative $\frac{dP}{dD}$.

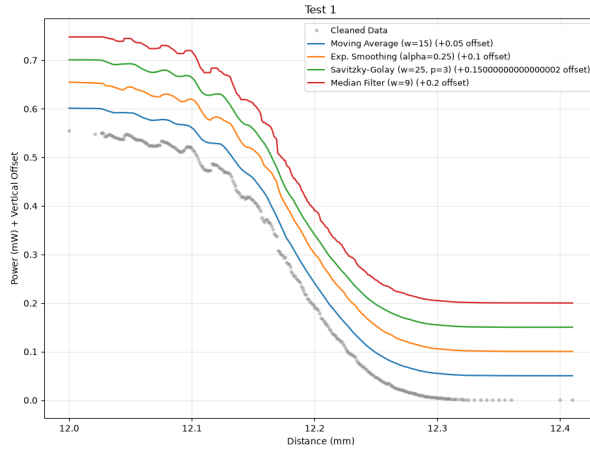


Figure 8: Test 1: Smoothed power curves (offset for clarity).

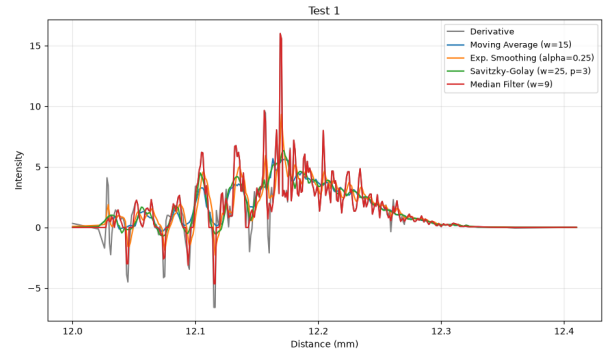


Figure 9: Test 1: Derivatives of standard smoothing outputs.

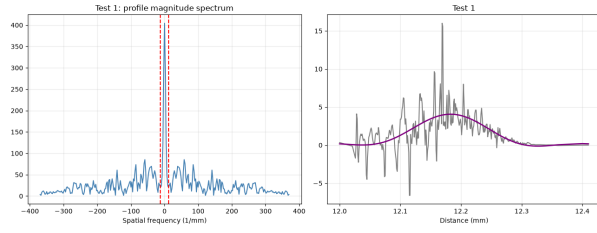


Figure 10: Test 1: Magnitude spectrum and spatial FFT filter.

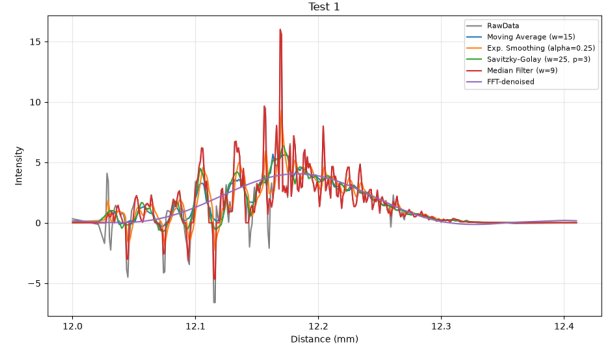


Figure 11: Test 1: FFT-denoised signal vs. standard smoothing.

Test 2:

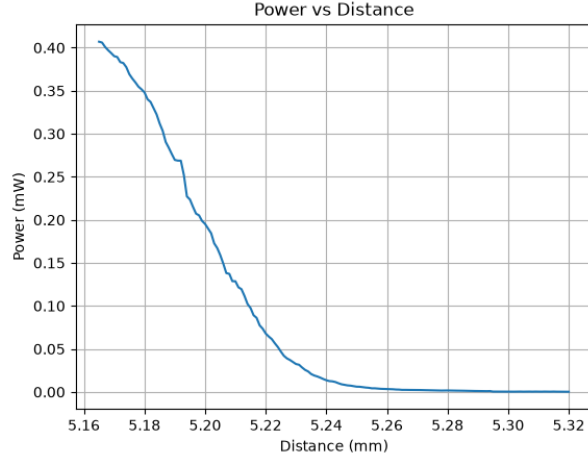


Figure 12: Test 2: Raw Power vs. Distance.

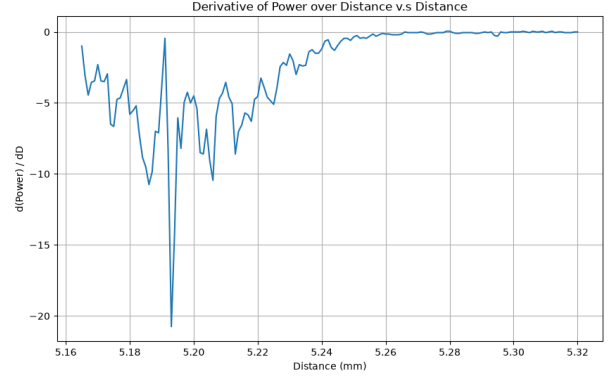


Figure 13: Test 2: Raw spatial derivative $\frac{dP}{dD}$.

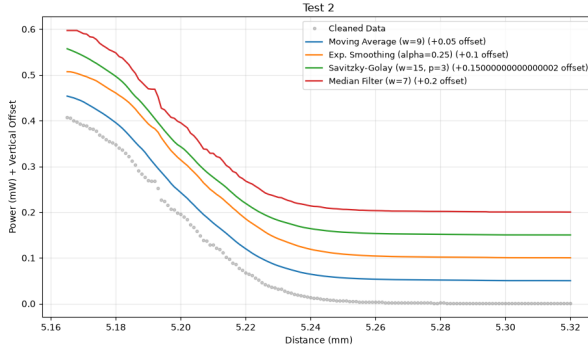


Figure 14: Test 2: Smoothed power curves (offset for clarity).

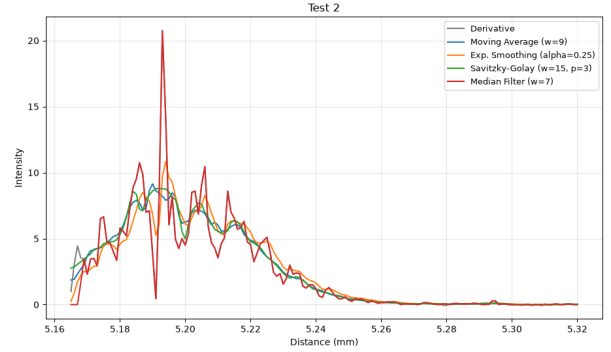


Figure 15: Test 2: Derivatives of standard smoothing outputs.

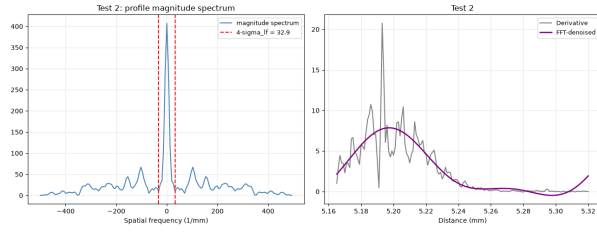


Figure 16: Test 2: Magnitude spectrum and spatial FFT filter.

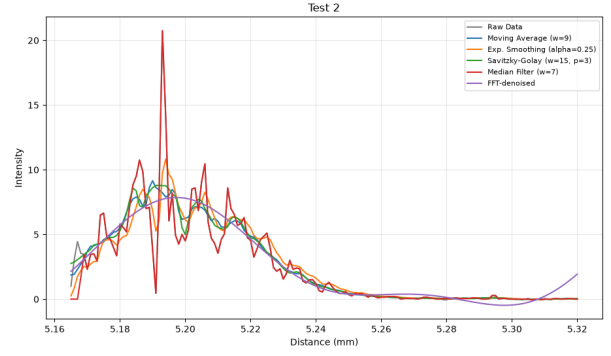
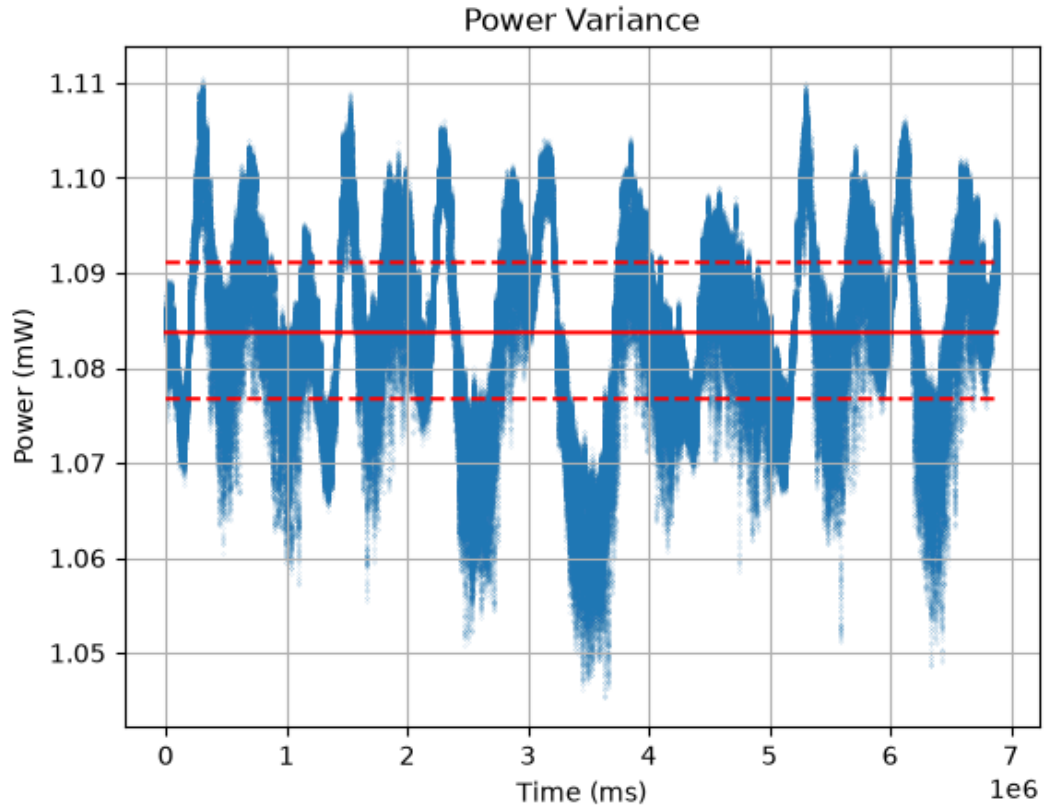


Figure 17: Test 2: FFT-denoised signal vs. standard smoothing.

6 Error Analysis

Count	661241
Mean	1.083723
Std Dev	0.010246
Min	1.045410
Max	1.110632
25%	1.076742
50%	1.084158
75%	1.091096

Table 5: Statistical Summary of Ambient Beam Power



7 Operating Instructions

7.1 Prerequisites

- Ensure curtains are closed and laser indicator is on outside the lab.
- Take off all jewelry and don appropriate laser safety goggles.
- Verify the beam path is clear of obstructions and that all optical components are securely mounted.
- Verify the power meter is aligned and is properly rated for the expected power levels, default to 500 mW.
- Initiate appropriate music... good fall backs are Led Zeppelin, Black Sabbath, or AC/DC.

7.2 Diode Initialization

1. Turn on the diode, ilx, it is located on the top shelf on the far right, it is blue.
2. Check the current and temperature settings on the diode controller, Temp should be 19.5°C and Current should be 165 mA.
3. Turn on the temperature via TEC controler, push ON.
4. Turn on the current via the controller, push ON.

7.3 PPLN Initialization

1. Verify HCP is set to 59.5°C, then turn on the PPLN temperature controller.

PUT ON SAFETY GOGGLES BEFORE PROCEEDING FURTHER

7.4 Amplifier Initialization

1. Turn on the amplifier, but leave the key switched off for now.
2. Verify the amplifier settings: the current should be set to 20 mA.
3. Turn on the amplifier key switch, then wait for system to boot through serial number and loading screens.
4. Turn on the amplifier output by turning the dial to highlight *START*, then push the dial in.

7.5 How to Shut Down the Laser System

1. Turn the key to off on the amplifier and then shut it down.
2. Flip the current off and then the TEC peltier device off.
3. Turn off the crystal heater

7.6 Troubleshooting

Placeholder for troubleshooting instructions.

8 Optical Components List

Part #	Specs	Description
Thor Labs Bin #1: Lenses		
LA1951	$f = 25.4$ mm	PCX $\varnothing 25.4$ mm
LA1805	$f = 30$ mm	PCX $\varnothing 25.4$ mm
LA1027	$f = 35$ mm	PCX $\varnothing 25.4$ mm
LA1422	$f = 40$ mm	PCX $\varnothing 25.4$ mm
LA1131	$f = 50$ mm	PCX $\varnothing 25.4$ mm
LA1134	$f = 60$ mm	PCX $\varnothing 25.4$ mm
LA1608	$f = 75$ mm	PCX $\varnothing 25.4$ mm
LA1509	$f = 100$ mm	PCX $\varnothing 25.4$ mm
LA1986	$f = 125$ mm	PCX $\varnothing 25.4$ mm
LA1433	$f = 150$ mm	PCX $\varnothing 25.4$ mm
LA1229	$f = 175$ mm	PCX $\varnothing 25.4$ mm
LA1708	$f = 200$ mm	PCX $\varnothing 25.4$ mm
LA1461	$f = 250$ mm	PCX $\varnothing 25.4$ mm
LA1484	$f = 300$ mm	PCX $\varnothing 25.4$ mm
LA1172	$f = 400$ mm	PCX $\varnothing 25.4$ mm
LA1908	$f = 500$ mm	PCX $\varnothing 25.4$ mm
LA1978	$f = 750$ mm	PCX $\varnothing 25.4$ mm
LA1464	$f = 1000$ mm	PCX $\varnothing 25.4$ mm
LB1761	$f = 25.4$ mm	BCX $\varnothing 25.4$ mm
LB1757	$f = 30$ mm	BCX $\varnothing 25.4$ mm
LB1811	$f = 35$ mm	BCX $\varnothing 25.4$ mm
LB1027	$f = 40$ mm	BCX $\varnothing 25.4$ mm
LB1471	$f = 50$ mm	BCX $\varnothing 25.4$ mm
LB1596	$f = 60$ mm	BCX $\varnothing 25.4$ mm
LB1901	$f = 75$ mm	BCX $\varnothing 25.4$ mm
LB1676	$f = 100$ mm	BCX $\varnothing 25.4$ mm
LB1904	$f = 125$ mm	BCX $\varnothing 25.4$ mm
LB1437	$f = 150$ mm	BCX $\varnothing 25.4$ mm
LB1294	$f = 175$ mm	BCX $\varnothing 25.4$ mm
LB1945	$f = 200$ mm	BCX $\varnothing 25.4$ mm
LB1056	$f = 250$ mm	BCX $\varnothing 25.4$ mm
LB1779	$f = 300$ mm	BCX $\varnothing 25.4$ mm
LB1391	$f = 400$ mm	BCX $\varnothing 25.4$ mm
LB1869	$f = 500$ mm	BCX $\varnothing 25.4$ mm
LB1475	$f = 750$ mm	BCX $\varnothing 25.4$ mm
LB1409	$f = 1000$ mm	BCX $\varnothing 25.4$ mm
LC1715	$f = -50$ mm	PCV $\varnothing 25.4$ mm
LC1582	$f = -75$ mm	PCV $\varnothing 25.4$ mm
LC1120	$f = -100$ mm	PCV $\varnothing 25.4$ mm
LD1464	$f = -50$ mm	BCV $\varnothing 25.4$ mm
LD1170	$f = -75$ mm	BCV $\varnothing 25.4$ mm

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Table 6 – continued from previous page

Part #	Specifications	Description
LD1613	$f = -100$ mm	BCV $\varnothing 25.4$ mm
LE	—	MP $\varnothing 25.4$ mm
LE1156	$f = 125$ mm	MP $\varnothing 25.4$ mm
LE1104	$f = 150$ mm	MP $\varnothing 25.4$ mm
LE1202	$f = 200$ mm	MP $\varnothing 25.4$ mm
LE1157	$f = 250$ mm	MP $\varnothing 25.4$ mm
LE1929	$f = 300$ mm	MP $\varnothing 25.4$ mm
LE1872	$f = 400$ mm	MP $\varnothing 25.4$ mm
LE1261	$f = 500$ mm	MP $\varnothing 25.4$ mm
LE1458	$f = 1000$ mm	MP $\varnothing 25.4$ mm
LF1822	$f = -100$ mm	MN $\varnothing 25.4$ mm
LF1510	$f = -125$ mm	MN $\varnothing 25.4$ mm
LF1547	$f = -150$ mm	MN $\varnothing 25.4$ mm
LF1097	$f = -200$ mm	MN $\varnothing 25.4$ mm
LF1774	$f = -250$ mm	MN $\varnothing 25.4$ mm
LF1015	$f = -300$ mm	MN $\varnothing 25.4$ mm
LF1544	$f = -400$ mm	MN $\varnothing 25.4$ mm
LF1988	$f = -500$ mm	MN $\varnothing 25.4$ mm
LF1141	$f = -1000$ mm	MN $\varnothing 25.4$ mm
Thor Labs Bin #2: Mounting Bases & Post Holders		
PH1	1" Long	$\varnothing 1/2''$ Post Holder
PH1.5	1.5" Long	$\varnothing 1/2''$ Post Holder
PH2	2" Long	$\varnothing 1/2''$ Post Holder
PH3	3" Long	$\varnothing 1/2''$ Post Holder
PH4	4" Long	$\varnothing 1/2''$ Post Holder
PH6	6" Long	$\varnothing 1/2''$ Post Holder
CL5	—	Standard Base Clamp
BA1S	$1'' \times 2.3'' \times 3/8''$	Mounting Base
BA1	$1'' \times 3'' \times 3/8''$	Mounting Base
BA2	$2'' \times 3'' \times 3/8''$	Mounting Base
Thor Labs Bin #3: Posts, Clamps, & Accessories		
TR075	$L = 0.75''$	$\varnothing 1/2''$ Stainless Steel Posts
TR1	$L = 1''$	$\varnothing 1/2''$ Stainless Steel Posts
TRK05	Alignment Key	$\varnothing 1/2''$ Stainless Steel Posts
TR1.5	$L = 1.5''$	$\varnothing 1/2''$ Stainless Steel Posts
TR2	$L = 2''$	$\varnothing 1/2''$ Stainless Steel Posts
TR3	$L = 3''$	$\varnothing 1/2''$ Stainless Steel Posts
TR4	$L = 4''$	$\varnothing 1/2''$ Stainless Steel Posts
TR6	$L = 6''$	$\varnothing 1/2''$ Stainless Steel Posts
TR8	$L = 8''$	$\varnothing 1/2''$ Stainless Steel Posts
TRA2	$L = 2''$	$\varnothing 1/2''$ Aluminum Posts
TRA3	$L = 3''$	$\varnothing 1/2''$ Aluminum Posts

Continued on next page

Table 6 – continued from previous page

Part #	Specifications	Description
R2	0.44" Thick	Slip-On Collars for $\varnothing 1/2''$ Posts
R2T	0.23" Thick	Slip-On Collars for $\varnothing 1/2''$ Posts
PMTR	Component Clamp	Slip-On Collars for $\varnothing 1/2''$ Posts
TR1M	1 mm Thick	Spacers for $\varnothing 1/2''$ Posts
TR3M	3 mm Thick	Spacers for $\varnothing 1/2''$ Posts
TR5M	5 mm Thick	Spacers for $\varnothing 1/2''$ Posts
TR10M	10 mm Thick	Spacers for $\varnothing 1/2''$ Posts
RA90	Right-Angle Clamp	Clamps and Mounts for $\varnothing 1/2''$ Posts
RA180	Right-Angle End Clamp	Clamps and Mounts for $\varnothing 1/2''$ Posts
SWC	Swivel Clamp	Clamps and Mounts for $\varnothing 1/2''$ Posts
TRB1	Ball and Socket Mount (8-32)	Clamps and Mounts for $\varnothing 1/2''$ Posts
Thor Labs Bin #4: SM1 Lens Tubes & Optomechanics		
SM1L03	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L05	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L10	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L15	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L20	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L30	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L40	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1E60	—	Extension Tube for $\varnothing 1''$ Optics
SM1M05	—	SM1 Lens Tubes Without External Threads
SM1M10	—	SM1 Lens Tubes Without External Threads
SM1M20	—	SM1 Lens Tubes Without External Threads
SM1M30	—	SM1 Lens Tubes Without External Threads
SM1S10	—	SM1 Lens Tube Spacers
SM1S20	—	SM1 Lens Tube Spacers
SM1L20C	—	SM1 Slotted Lens Tubes
SM1L30C	—	SM1 Slotted Lens Tubes
SM1V05	—	Adjustable SM1 Lens Tubes
SM1V10	—	Adjustable SM1 Lens Tubes
SM1V15	—	Adjustable SM1 Lens Tubes
SM1ZM	—	SM1 Zoom Housings
??	—	SM1 Zoom Housings
AD9.5F	—	SM1-Threaded Adapters for Cylindrical Components
AD11F	—	SM1-Threaded Adapters for Cylindrical Components
AD12F	—	SM1-Threaded Adapters for Cylindrical Components
AD15F	—	SM1-Threaded Adapters for Cylindrical Components
SM1SH1	—	SM1 Beam Shutter
SM1D12D	—	SM1 Iris Diaphragm

Continued on next page

Table 6 – continued from previous page

Part #	Specifications	Description
SM1B3	—	SM1 Lens Tube Bellows
SM1TC	—	Clamp for SM1 Lens Tubes
SM1PL	—	SM1 Plug
SM1CP2	—	SM1 End Caps
SM1CP1	—	SM1 End Caps
VRC5M1	—	Alignment Disks
VRC2SM1	—	Alignment Disks
VRC4SM1	—	Alignment Disks
SM1T1	—	SM1 Lens Tube Couplers
SM1T2	—	SM1 Lens Tube Couplers
SM1T10	—	SM1 Lens Tube Couplers
SM1NT1	—	SM1 Slotted Locking Ring
SPW502	—	Spanner Wrenches
SPW602	—	Spanner Wrenches
SM1RR	—	Retaining Rings
SM1LTRR	—	Retaining Rings
Newport MKS Bin		
LH-0.5A	$\varnothing 0.5''$, 6 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-50R25	$\varnothing 0.25''$ Adapter, 3.5 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-1A	$\varnothing 1''$, 7 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-f.5A	$1.5''$, 9 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-30M	$\varnothing 30$ mm, 7 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-2A	$\varnothing 2''$, 9 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-3A	$\varnothing 3''$, 9 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-05N	$\varnothing 0.5''$, 7 mm Max Edge	A-Line Lens Mounts, Insertable
LH-1N	$\varnothing 1''$, 10 mm Max Edge	A-Line Lens Mounts, Insertable
LH-2N	$\varnothing 2''$, 13 mm Max Edge	A-Line Lens Mounts, Insertable
LH1-ATP	Alignment Target	A-Line Accessories, 8-32 Thread
LH-K1	$\varnothing 0.5''$ to $\varnothing 1''$ Link Block	A-Line Accessories, 8-32 Thread
LH-K2	$\varnothing 1''$ to $\varnothing 2''$ Link Block	A-Line Accessories, 8-32 Thread
LH-GR-4	$L = 4''$	A-Line Guide Rods
LH-GR-8	$L = 8''$	A-Line Guide Rods
LH-GR-10	$L = 10''$	A-Line Guide Rods
LFTO-WH	$\varnothing 1''$ Retaining Ring	Spanner Wrenches
SW-OM	$\varnothing 0.25''$ to $\varnothing 2''$ Retaining Ring	Spanner Wrenches

Notes

¹Wikipedia entry: Second-harmonic generation (SHG), also known as frequency doubling, is the lowest-order wave-wave nonlinear interaction that occurs in various systems, including optical, radio, atmospheric, and magnetohydrodynamic systems. As a prototype behavior of waves, SHG is widely used, for example, in doubling laser frequencies. SHG was initially discovered as a nonlinear optical process in which two photons with the same frequency interact with a nonlinear material, are “combined”, and generate a new photon with twice the energy of the initial photons (equivalently, twice the frequency and half the wavelength), conserving the coherence of the excitation. It is a special case of two-photon sum-frequency generation, and more generally of harmonic generation.

²**Phase-matching:** For sum-frequency generation to occur efficiently, phase-matching conditions must be satisfied: $\hbar k_3 \approx \hbar k_1 + \hbar k_2$, where k_1, k_2, k_3 are the angular wavenumbers of the three waves travelling through the medium. (This equation resembles conservation of momentum.) The sum-frequency generation becomes more efficient as this condition is satisfied more precisely.

³In this calculation I assume the full beam diameter $2\omega_{01} = 0.56$ mm because the beam emerges from a collimator. This assumes the beam waist may be located a short distance away from the collimator face, as noted in [6]: “The purpose of a beam collimator is essentially to transform a strongly diverging light beam into a collimated beam, i.e., a beam where light propagates essentially only in one direction, and the beam divergence is weak. The output beam may have its beam waist close to the output aperture, or a mild focus somewhat away from it.”

⁴“In general, a good rule of thumb is that the spot size should be chosen such that the Rayleigh range is half the length of the crystal.” If $z_{02} = L/2$, then $b = 2z_{02} = L$, so $L/b = 1$.

⁵Note: 1.21 mRad was a measured value, substantially lower than the spec sheet value of approximately 5 mRad.

⁶The following calculations were made under the (incorrect) assumption of a waist radius of $45 \mu\text{m}$ rather than the updated $279.9 \mu\text{m}$ derived here.

⁷Periodically poled lithium niobate (PPLN) is a lithium niobate crystal that switches polarity in a periodic fashion and is used for achieving quasi-phase-matching in nonlinear optics.

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